STRUCTURAL BLUEPRINT

G+1 Farm Storage + Residence

Mithanpur Urf Naimtulla Nagar, Moradabad, Uttar Pradesh

Project Type	G+1 RCC Frame Building (Storage + Residence)
Plot Area	20ft x 25ft internal (21.5ft x 26.5ft external)
Ground Floor	Agricultural Storage (Godown) with 10ft Shutter
First Floor	1BHK Residential (Room + Kitchen + Bathroom)
Structural System	RCC Moment Resisting Frame (OMRF), 8 Columns
Plinth Height	3ft (0.91m) above NGL
Floor Height	12ft (3.66m) per floor
Total Height	28ft to roof slab, 31ft to parapet
Front Shade	6ft pure cantilever (no supports)
Seismic Zone	IV (Z = 0.24, Ah = 0.10)
Wind Speed	47 m/s (Moradabad)
Soil Type	Alluvial (Medium), SBC = 130 kN/m2
Concrete Grade	M20 (fck = 20 N/mm2)
Steel Grade	Fe500 / Fe500D (fy = 500 N/mm2)
Applicable Codes	IS 456:2000, IS 875 (Parts 1-5), IS 1893:2016, IS 13920:2016

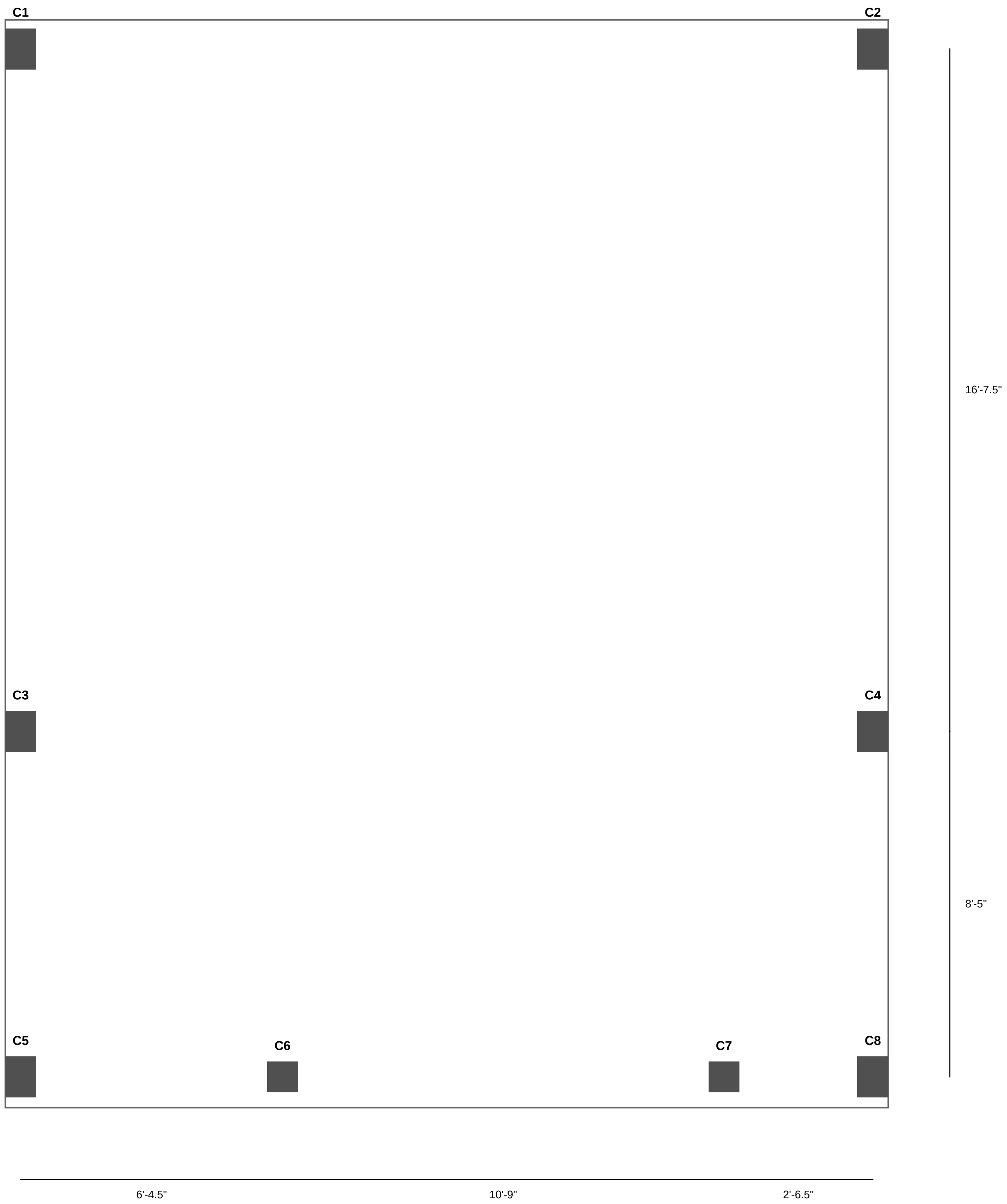
Date: June 2026

Drawing No: STR-001

Scale: NTS (Not to Scale) unless noted

SHEET 2: COLUMN GRID PLAN (As-Built Positions)

Plan View - Looking Down. Front = Bottom. All dimensions center-to-center.



External: 21'-6" x 26'-6" | Internal: 20'-0" x 25'-0"

COLUMN SIZES:

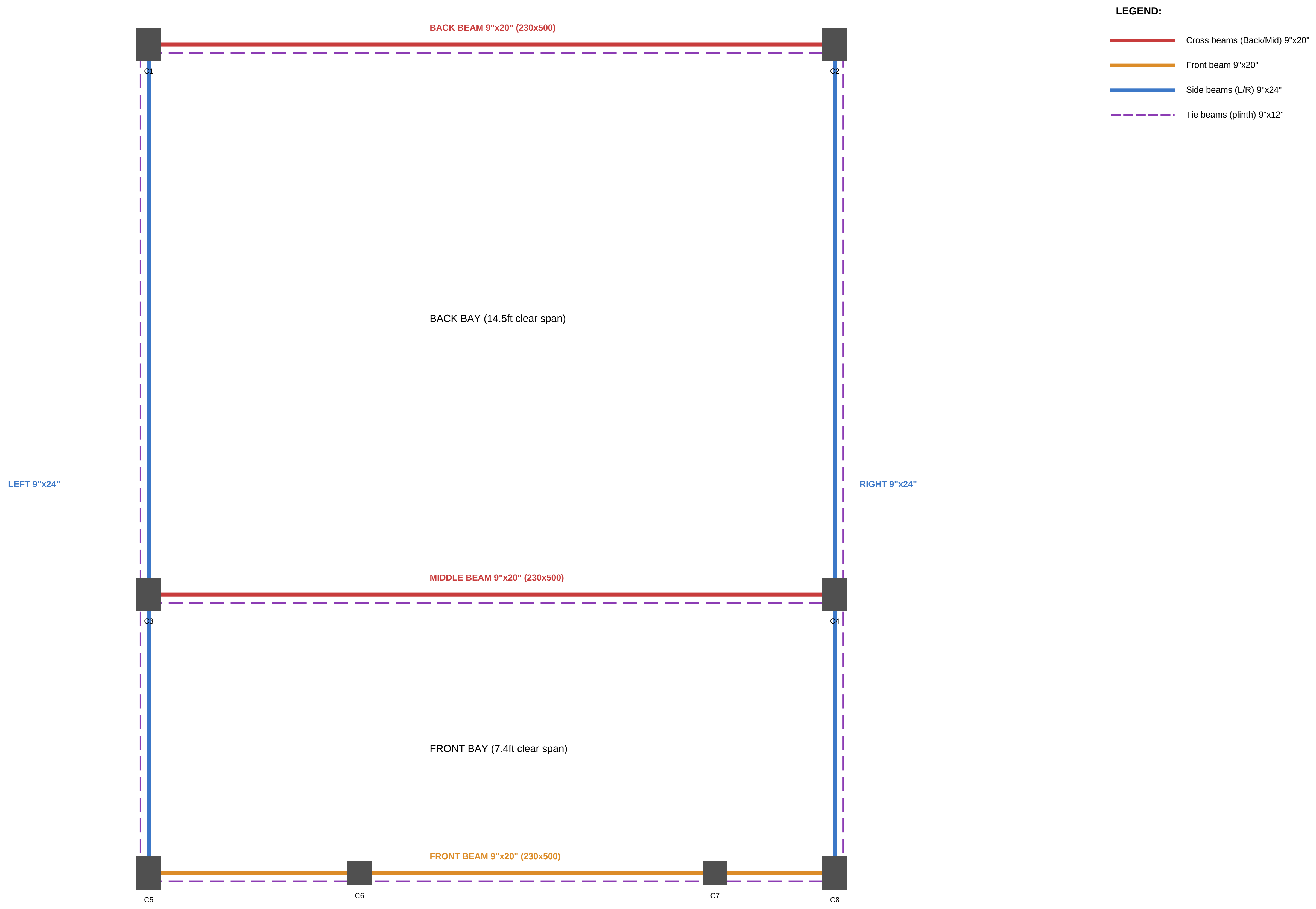
C1 to C5, C8 = 9" x 12" (230x300mm)

C6, C7 = 9" x 9" (230x230mm)

C6, C7 are GF only (do not extend to 1F)

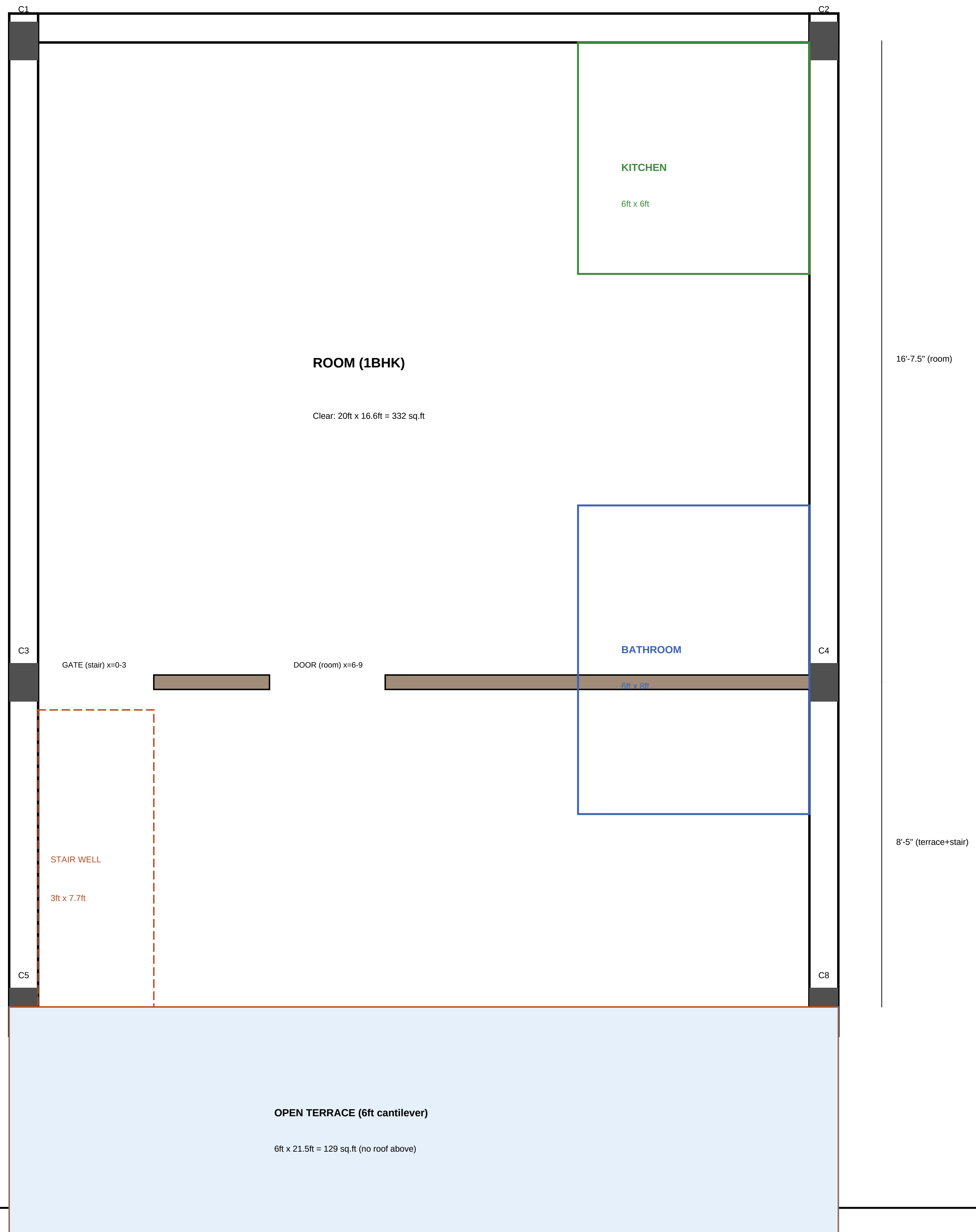
SHEET 3: BEAM LAYOUT PLAN

GF Ceiling Level (z = 15ft). Main beams + Tie beams at plinth level.



SHEET 4: FIRST FLOOR PLAN

Plan at z = 15.5ft (1F floor level). Shows room layout, partition, stairwell, kitchen, bathroom.

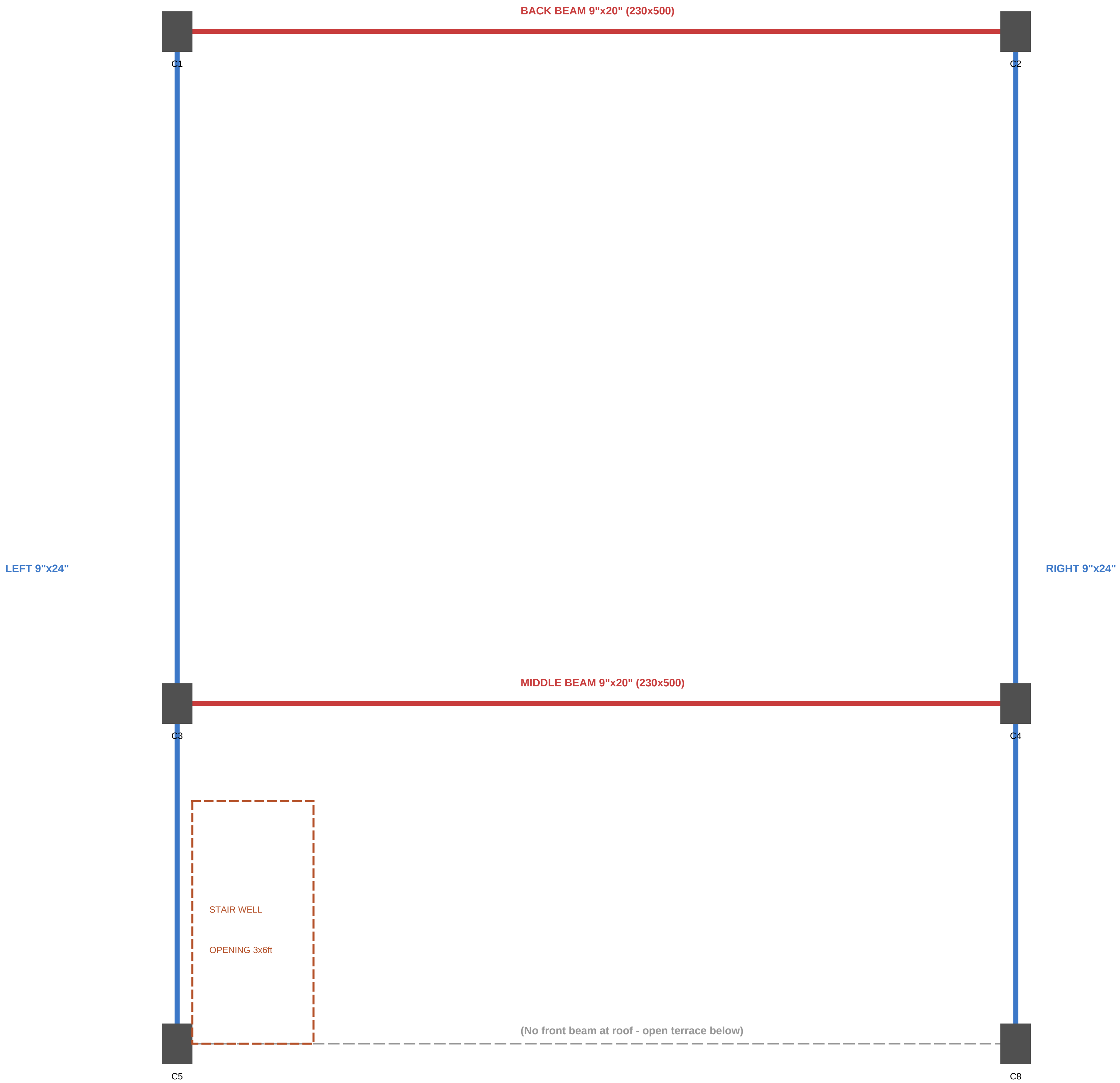


1F NOTES:

C6 & C7 do NOT extend to 1F
No front wall (open terrace)
Partition wall: 4.5" brick (115mm)
Floor: 6" RCC slab (from GF beams)
Roof: 6" RCC slab at z=27.5ft
Parapet: 4.5" brick, 3ft high
Terrace: cantilever, no roof
Stairwell: 3ft x 7.7ft opening
Sloped stair covers stairwell

SHEET 5: 1F ROOF BEAM LAYOUT (z = 27.5ft)

Same configuration as GF ceiling beams. Beams at roof level supporting roof slab + parapet.



1F ROOF BEAMS:

Same sizes as GF ceiling level beams.

Back + Middle: 9"x20" (230x500mm)

Left + Right: 9"x24" (230x600mm)

No front beam (1F has no front wall)

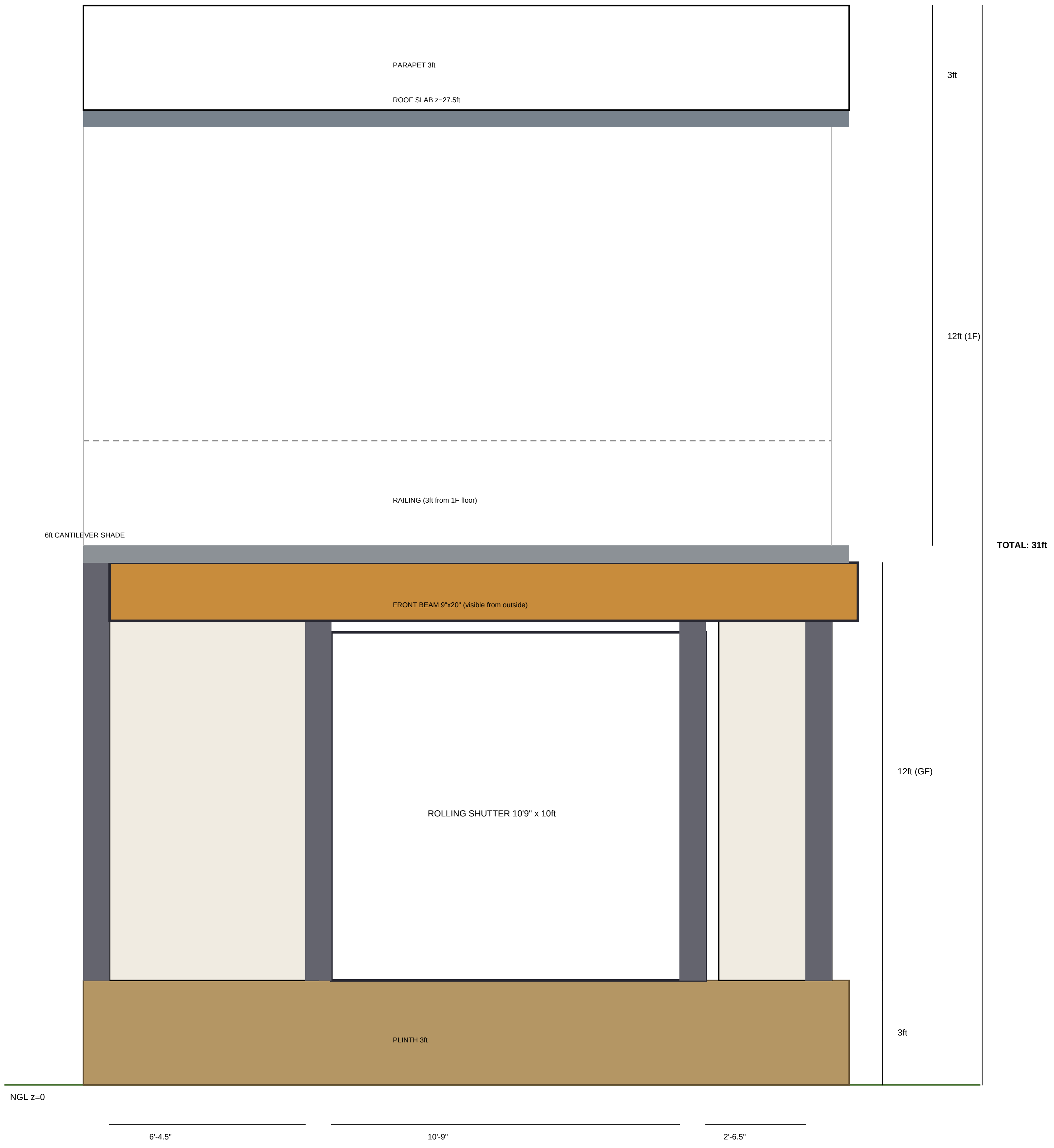
Roof slab: 150mm (6") RCC, same as floor

Parapet: 4.5" brick, 3ft high on all 3 sides

Stairwell opening: 3ft x 6ft (for sloped stair)

SHEET 6: FRONT ELEVATION

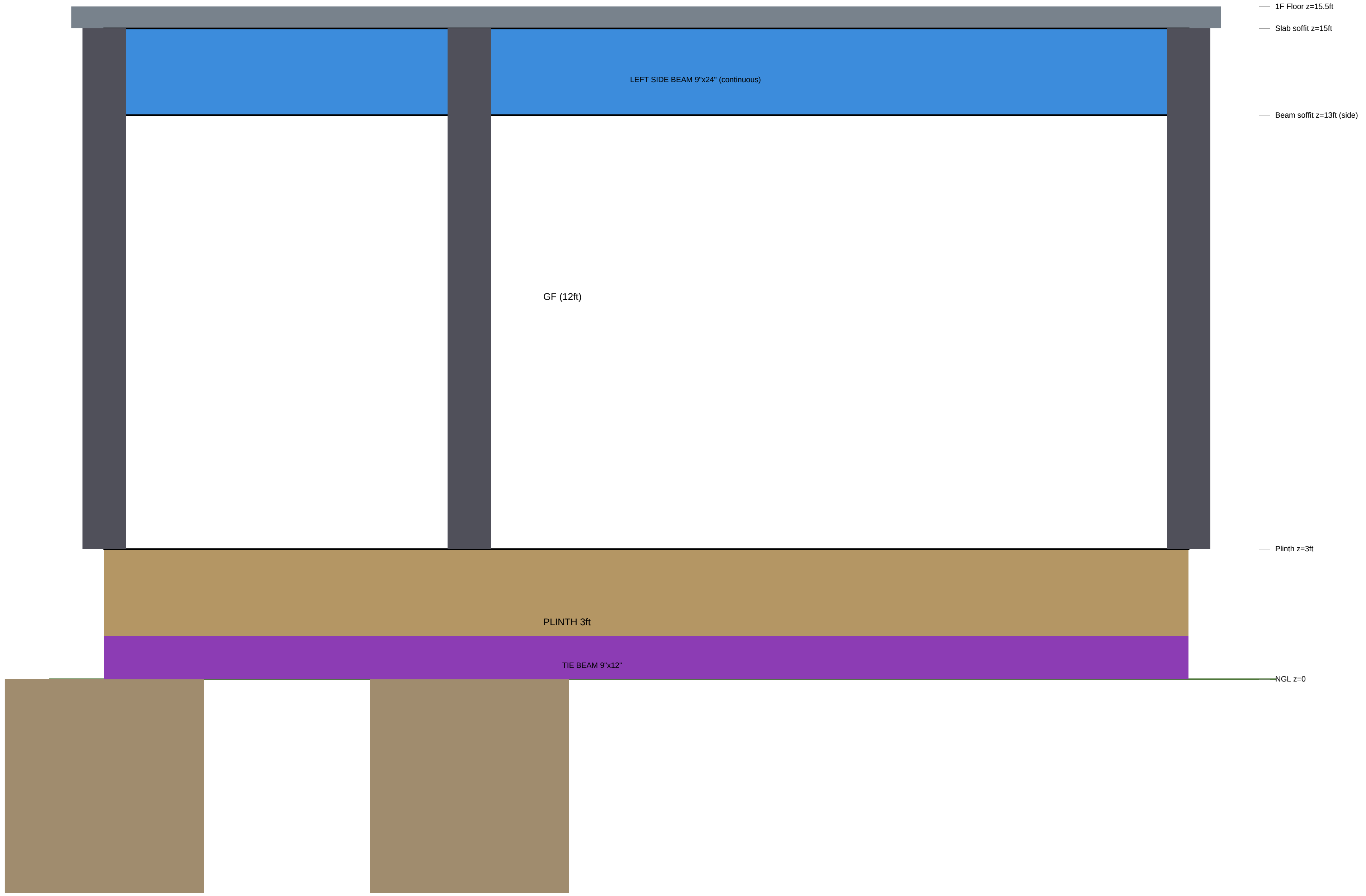
View from OUTSIDE looking at front face. Left = C5 side, Right = C8 side.



SHEET 7: STRUCTURAL SECTIONS

Section A-A (Front to Back, through left wall) | Section B-B (Left to Right, through middle beam)

SECTION A-A (Front to Back)



SHEET 6: COLUMN & TIE BEAM SCHEDULE

A. COLUMN SCHEDULE

Col	Size (mm)	Position	Main Steel	Ties	Height	Util.
C1	230x300 (9"x12")	Back-Left corner	4-16+4-12 (1.82%)	8mm@150/100	GF+1F (24ft)	45%
C2	230x300 (9"x12")	Back-Right corner	4-16+4-12 (1.82%)	8mm@150/100	GF+1F (24ft)	45%
C3	230x300 (9"x12")	Mid-Left (y=8'5")	6-16+2-12 (2.08%)	8mm@150/100	GF+1F (24ft)	52%
C4	230x300 (9"x12")	Mid-Right (y=8'5")	6-16+2-12 (2.08%)	8mm@150/100	GF+1F (24ft)	52%
C5	230x300 (9"x12")	Front-Left corner	4-16+4-12 (1.82%)	8mm@150/100	GF+1F (24ft)	45%
C6	230x230 (9"x9")	Front @ x=6'0"	4-12mm (0.85%)	8mm@150/100	GF only (12ft)	25%
C7	230x230 (9"x9")	Front @ x=16'9"	4-12mm (0.85%)	8mm@150/100	GF only (12ft)	25%
C8	230x300 (9"x12")	Front-Right corner	4-16+4-12 (1.82%)	8mm@150/100	GF+1F (24ft)	45%

B. TIE BEAM SCHEDULE (Plinth Level)

Tie Beam	Size (mm)	Span	Main Steel	Stirrups	Level
TB-Back (C1-C2)	230x300 (9"x12")	20.75ft/6.3m	4-12mm (2T+2B)	8mm @ 200 c/c	Plinth (z=0-1ft)
TB-Middle (C3-C4)	230x300 (9"x12")	20.75ft/6.3m	4-12mm (2T+2B)	8mm @ 200 c/c	Plinth (z=0-1ft)
TB-Front (C5-C8)	230x300 (9"x12")	20.75ft/6.3m	4-12mm (2T+2B)	8mm @ 150 c/c	Plinth (z=0-1ft)
TB-Left (C5-C3-C1)	230x300 (9"x12")	24.75ft/7.5m	6-12mm (3T+3B)	8mm @ 150 c/c	Plinth (z=0-1ft)
TB-Right (C8-C4-C2)	230x300 (9"x12")	24.75ft/7.5m	6-12mm (3T+3B)	8mm @ 150 c/c	Plinth (z=0-1ft)

NOTES:

1. Special confining zone (Lo) = 650mm from each beam-column joint face (IS 13920)
2. Stirrup spacing within Lo = 100mm c/c (8mm dia, rectangular)
3. No lap splices within joint zone. All column laps at MID-HEIGHT only.
4. Lap length = 50d (50 x 16 = 800mm for 16mm bars, 50 x 12 = 600mm for 12mm bars)
5. Column capacity (min steel): 815 kN. Max demand: 363 kN. Factor of safety: 2.25x

SHEET 7: BEAM SCHEDULE (GF Ceiling / 1F Floor Level)

A. MAIN BEAM SCHEDULE

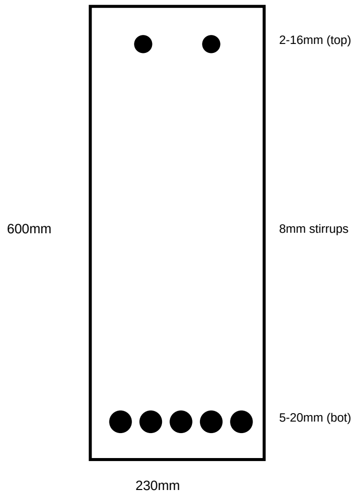
Beam	Size(mm)	Span	Bottom Steel	Top Steel	Stirrups	Mu(kNm)
Back (C1-C2)	230x500 (9"x20")	20.75ft/6.3m	4-20mm+2-16mm	4-20mm+2-16mm	8mm@100/150	266.3
Middle (C3-C4)	230x500 (9"x20")	20.75ft/6.3m	4-20mm+2-16mm	4-20mm+2-16mm	8mm@100/150	266.3
Front (C5-C8)	230x500 (9"x20")	20.75ft/6.3m*	3-20mm+2-16mm	2-20mm+2-16mm	8mm@100/150	143.8
Left (C1-C3-C5)	230x600 (9"x24")	24.75ft/7.5m	5-20mm	2-16+2-20(sup)	8mm@100/175	386.1
Right (C2-C4-C8)	230x600 (9"x24")	24.75ft/7.5m	5-20mm	2-16+2-20(sup)	8mm@100/175	386.1

* Front beam segmented by C6, C7. Max segment span = 10.75ft (C6-C7, shutter opening).

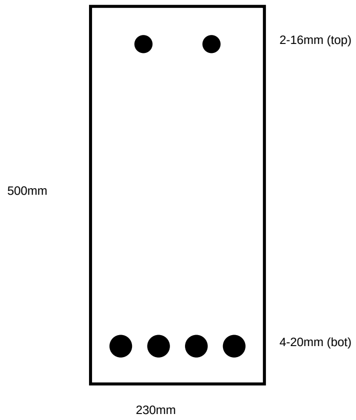
Stirrups: @100mm c/c within L/4 from each support; @150-175mm c/c at midspan.

B. BEAM CROSS-SECTIONS (Typical)

9"x24" SIDE BEAM



9"x20" CROSS BEAM



SHEET 8: SLAB & FOUNDATION DETAILS

A. SLAB SCHEDULE

Slab	Thick.	Type	Bottom Steel	Top Steel	Span
1F Floor (main)	150mm (6")	Two-way	10mm@150 c/c both	8mm@200 at supports	20'x25'/6.1x7.6m
Roof slab	150mm (6")	Two-way	10mm@150 c/c both	8mm@200 at supports	20'x25'/6.1x7.6m
Cantilever (front)	150mm (6")	Cantilever	8mm@200 (bottom)	12mm@125 (TOP-main)	6ft/1.83m proj.
GF floor (on earth)	100mm (4")	On grade	6mm mesh@150	N/A	On soil
Staircase waist	125mm (5")	One-way	12mm@150 c/c	8mm@200 (distr.)	~8ft/2.5m

B. FOUNDATION SCHEDULE

Footing For	Size	Depth	Thickness	Steel	SBC	Type
Interior (C3,C4)	1.4x1.4m (4.6'x4.6')	1.5m/5ft BGL	300mm/12"	12mm@150 both	130 kN/m2	Isolated
Corner (C1,C2,C5,C8)	1.4x1.5m (4.6'x5')	1.5m/5ft BGL	300mm/12"	12mm@150 both	130 kN/m2	Isolated
Front (C6,C7)	1.2x1.2m (4'x4')	1.5m/5ft BGL	250mm/10"	10mm@150 both	130 kN/m2	Isolated

All footings connected by tie beams (230x300mm) at plinth level. PCC bedding: 150mm M10 below all footings.

SHEET 9: GENERAL CONSTRUCTION NOTES & SPECIFICATIONS

A. MATERIALS

Concrete: M20 grade (1:1.5:3 by volume) for all RCC work. OPC 43/53 grade cement.

Steel: Fe500 / Fe500D TMT bars. BIS certified (IS 1786). No re-rolled or plain MS bars.

Bricks: First class burnt clay bricks (compressive strength > 7.5 N/mm2).

Sand: Clean river sand (Zone II as per IS 383). Free from clay, silt, organic matter.

Coarse aggregate: 20mm nominal size (angular crushed stone). Clean, well-graded.

Water: Potable quality. No sea water or contaminated water for mixing or curing.

B. CLEAR COVER TO REINFORCEMENT

Foundations: 50mm/2" (bottom), 40mm/1.5" (sides)

Columns: 40mm/1.5" all faces

Beams: 25mm/1" (bottom and sides)

Slabs: 20mm/3/4" (bottom), 15mm/5/8" (top)

Exposed surfaces: 50mm/2" minimum

C. LAP LENGTHS & ANCHORAGE (Seismic Zone IV)

Tension lap (columns/beams): 50d (50 x bar dia). E.g., 20mm bar = 1000mm/3'4" lap.

Compression lap: 40d. E.g., 16mm bar = 640mm/2'1" lap.

Column laps: ONLY at mid-height (6ft from floor). Never in the joint zone (Lo = 650mm/2'2").

Beam bars: Must extend minimum Ld (development length) past the face of support.

Top bars in cantilever: Anchor 10ft/3m into the main span. Critical for safety.

D. SEISMIC DETAILING (IS 13920)

All beam-column joints: Special confining reinforcement required.

Stirrup spacing in confining zone (Lo): max d/4 or 100mm (whichever is less).

Confining zone length (Lo): max(D, h/6, 450mm) = 650mm from each joint face.

Bottom bars at beam support: Min 50% of top bar area must continue through joint.

No hooks opened out. All stirrups: 135-degree hooks with 10d extension.

E. CURING & CONSTRUCTION SEQUENCE

Curing: Minimum 14 days continuous water curing (ponding for slabs, wet gunny for columns).

Formwork removal: Beams sides 3 days, slab props 21 days, cantilever props 28 days.

Sequence: Foundation > Plinth beams > Plinth fill > GF floor > Columns > Beams+Slab (together).

Concrete must be placed within 30 min of mixing. No re-tempering with water.

Vibration: Needle vibrator mandatory for all beams and columns. No hand-tamping.

SHEET 10: INDEPENDENT STRUCTURAL REVIEW & RECOMMENDATIONS

A. OVERALL STRUCTURAL CONFIDENCE ASSESSMENT

This structure has been reviewed against IS 456:2000, IS 1893:2016, and IS 13920:2016 for a G+1 RCC frame building in Seismic Zone IV. The 8-column moment-resisting frame (OMRF, R=3.0) with as-built column positions is assessed for gravity loads, seismic forces, and serviceability.

B. ELEMENT-WISE ADEQUACY RATING

Element	Rating	Basis	Concern (if any)
Columns (230x300)	ADEQUATE	45-52% utilization, FoS 2.25	Strong col-weak beam marginal
Side beams (230x600)	ADEQUATE	Mu,cap > Mu,demand (doubly reinf.)	At design limit for 7.62m span
Cross beams (230x500)	ADEQUATE	Doubly reinforced design	Heavy at 1F level (266 kNm)
Front beam (230x500)	ADEQUATE	Segmented by C6/C7	Shutter span unsupported below
Tie beams (230x300)	ADEQUATE	Nominal ties for seismic	L/R tie beam: 15.5ft span is long
Floor slab (150mm)	MARGINAL	OK for strength	Back bay deflection borderline
Cantilever (150mm, 6ft)	MARGINAL	Strength OK	L/d ratio 14.6 > 9.8 limit
Foundations (1.4x1.4m)	ADEQUATE	SBC utilization ~60%	Needs soil test confirmation
Seismic drift (Y-dir)	MARGINAL	Amplified drift ~39mm	Exceeds limit; infill helps

C. RECOMMENDATIONS FOR IMPROVEMENT (Without Major Rework)

1. OPTIONAL PILLAR at x=7'1.5", y=8'5" (near middle beam, left of C6): Provides direct load path for staircase reactions into the middle beam. Reduces beam bending at that point. Size: 9"x9" (230x230mm), 4-12mm steel, same height as GF columns. Low cost, high benefit.

2. CANTILEVER EDGE BEAM: Add a 150x230mm upstand beam at the 6ft cantilever tip. Reduces deflection by ~40%. Alternatively, increase cantilever slab to 175mm. Cost: ~Rs 3,000 additional. Recommended for long-term crack prevention.

3. BACK BAY SLAB: Increase from 150mm to 175mm (7") for the back bay panel only (between middle beam and back beam). Resolves deflection concern (L/d improves to 28). Cost: ~Rs 12,000 additional concrete + steel. Strongly recommended.

4. INFILL MASONRY: The 9" brick walls (though non-structural) will significantly improve lateral stiffness and reduce seismic drift to well within limits. Ensure proper separation gaps (10-15mm) at beam-wall interface to prevent strut damage. OR mortar-fill for added stiffness.

5. SIDE BEAM MID-SPAN: If any deflection visible during construction, add MS angle bracket (100x100x10mm) as a knee brace from C3/C4 column face to beam soffit at mid-span.

D. PROFESSIONAL SIGN-OFF

STRUCTURAL ADEQUACY CERTIFICATE

Based on preliminary structural analysis per IS 456:2000, IS 875:1987/2015, IS 1893:2016, and IS 13920:2016, this G+1 RCC frame structure is found ADEQUATE for gravity and seismic loads with the noted recommendations. The structure is APPROVED FOR CONSTRUCTION subject to: (a) implementation of recommendations #2 and #3 above, (b) proper seismic detailing per IS 13920, and (c) final design verification by a licensed structural engineer before beam/slab casting.

Confidence Level: 85% (GOOD)   Risk Category: LOW-MODERATE